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CSE370 : Database Systems
Project Report
Project Title : Smart Blood Donor

Group No : 06, CSE370 Lab Section : 14, Spring 2026		
ID	**Name**	**Contribution**
24101406	Farhan Sadik	Database Development, Backend Development, Overall Testing
24101339	Kazi Zarif Yamin	Database Development, Frontend Development, Overall Testing
24101363	Syeda Fatima Tasnia	Database Development, Backend Development, Overall Debugging

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Introduction

The Smart Blood & Emergency Donor Network is a web-based database system designed to address a critical gap in blood donation management, connecting donors, recipients and blood banks in a fast, coordinated, and data-driven way. In emergency situations, finding the right donor at the right time can mean the difference between life and death. This system automates that process.

The platform supports three types of users: donors, recipients and administrators each with a dedicated dashboard and role-specific functionality. Donors register with their health details and blood type, manage their availability, respond to emergency match notifications and earn rewards through donation campaigns. Recipients can submit emergency blood requests, view matched donors and leave feedback after a donation. Administrators oversee the entire lifecycle: validating requests, triggering intelligent donor matching, managing blood bank inventory and moderating user accounts.

At the core of the system is an intelligent matching engine that evaluates donor eligibility based on multiple criteria like blood type compatibility, hemoglobin levels, weight thresholds and 90-day donation cooldown. Only donors who pass all conditions are surfaced to the recipient and notified. This ensures both safety and efficiency.

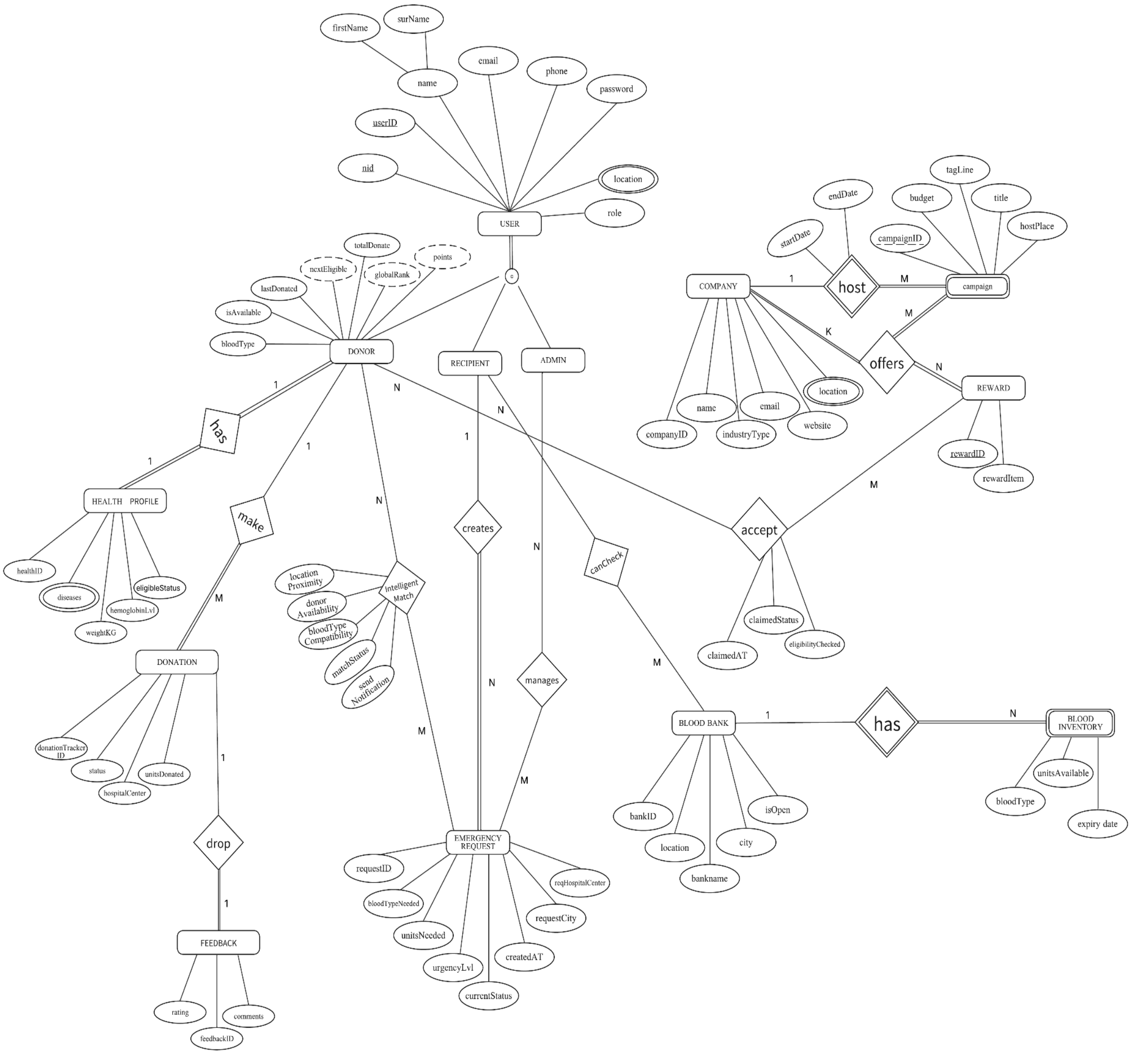
Beyond emergency handling, the system tracks blood inventory across multiple blood banks with expiry monitoring, runs company-sponsored donation campaigns with reward redemption, and maintains a donor leaderboard based on total donations. All features are backed by a normalized relational database schema designed following ER/EER modeling principles.

The frontend is built with PHP, HTML, CSS & the backend uses MySQL. Together, these components form a complete, functional, and deployable system for blood donation management.

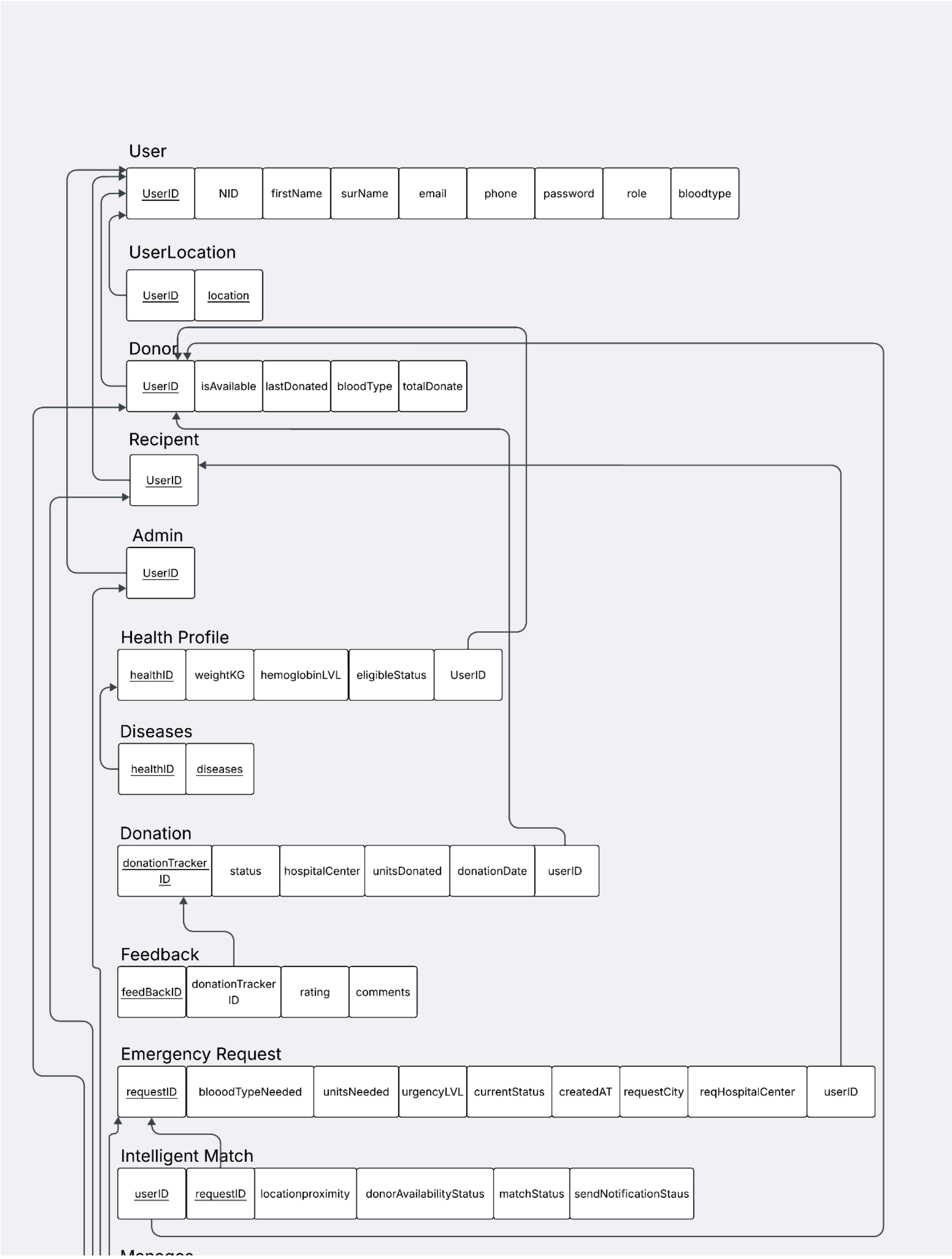
Project Features

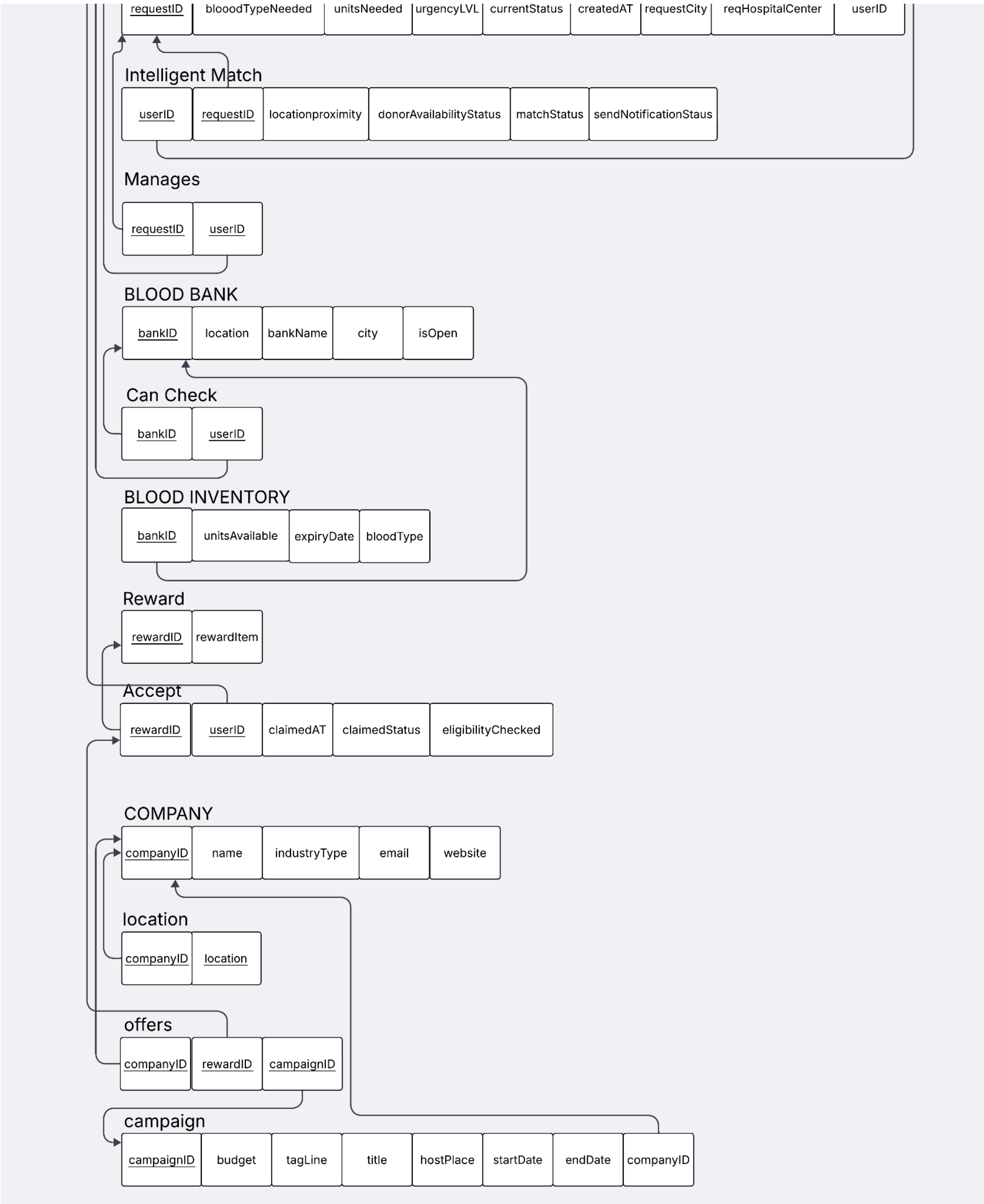
ID Name	Features	
24101406 Farhan Sadik	Ft 1	Emergency Request Processing Logic
	Ft 2	Donor Health & Points Profile Updates
	Ft 3	Find Donor Compatibility
24101339 Kazi Zarif Yamin	Ft 1	Feedback Data Operations
	Ft 2	Recipient Dashboard Data Retrieval & Validation
	Ft 3	Blood Bank Inventory Operations
24101363 Syeda Fatima Tasnia	Ft 1	Reward Claim Database Transactions
	Ft 2	Active Campaign Management Logic
	Ft 3	Admin Moderation Endpoints

ER/EER Diagram



Schema Diagram




requestID
bloodTypeNeeded
unitsNeeded
urgencyLVL
currentStatus
createdAT
requestCity
reqHospitalCenter
userID
Intelligent Match
userID
requestID
locationproximity
donorAvailabilityStatus
matchStatus
sendNotificationStaus
Manages
requestID
userID
BLOOD BANK
bankID
location
bankName
city
isOpen
Can Check
bankID
userID
BLOOD INVENTORY
bankID
unitsAvailable
expiryDate
bloodType
Reward
rewardID
rewardItem
Accept
rewardID
userID
claimedAT
claimedStatus
eligibilityChecked
COMPANY
companyID
name
industryType
email
website
location
companyID
location
offers
companyID
rewardID
campaignID
campaign
campaignID
budget
tagLine
title
hostPlace
startDate
endDate
companyID

Normalization

a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

Answer :

a) 1NF (First Normal Form)

Our converted schema is in 1NF because all attributes are atomic, meaning there are no multivalued or composite attributes and no repeating groups. Each field contains only a single value.

b) 2NF (Second Normal Form)

Our schema is in 2NF because it is already in 1NF and there is no partial dependency in our relations. Partial dependency usually happens when a table has a composite primary key and some non-key attributes depend on only part of that key. But, our schema satisfies 2NF and partial functional dependencies are not present.

c) 3NF (Third Normal Form)

Finally, the schema is also in Third Normal Form (3NF) because there are no transitive dependencies present, meaning that non-key attributes do not depend on other non-key attributes but only on the primary key. As a result, the database design avoids redundancy and update anomalies, confirming that it is properly structured and fully normalized up to 3NF.

Frontend Development

The frontend development was done using HTML,CSS & JS

Contribution of ID : 24101406, Name : Farhan Sadik

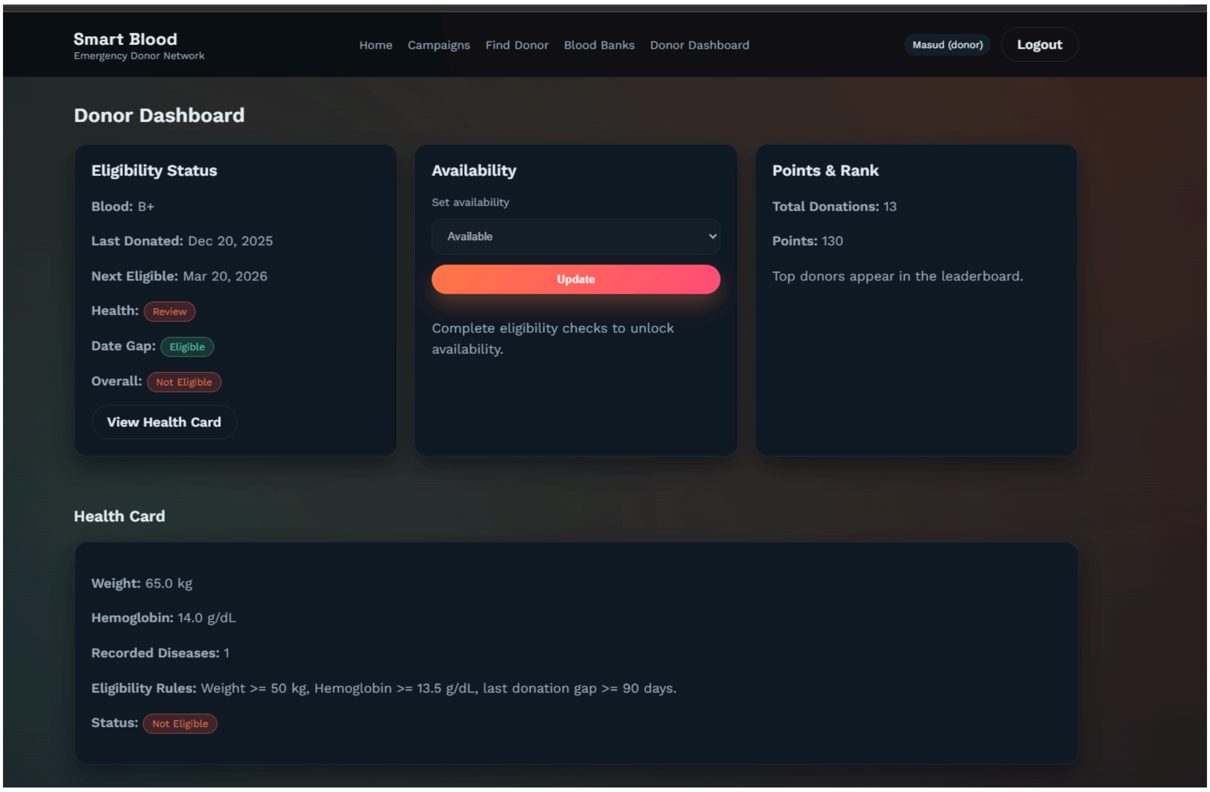
Designing and structuring complex, interactive user interface components, including dynamic emergency request forms, the donor dashboard (incorporating health metrics displays, points system counters, and availability toggle controls) and the find-donor compatibility and multi-conditional search interface, ensuring intuitive usability and responsive data presentation.

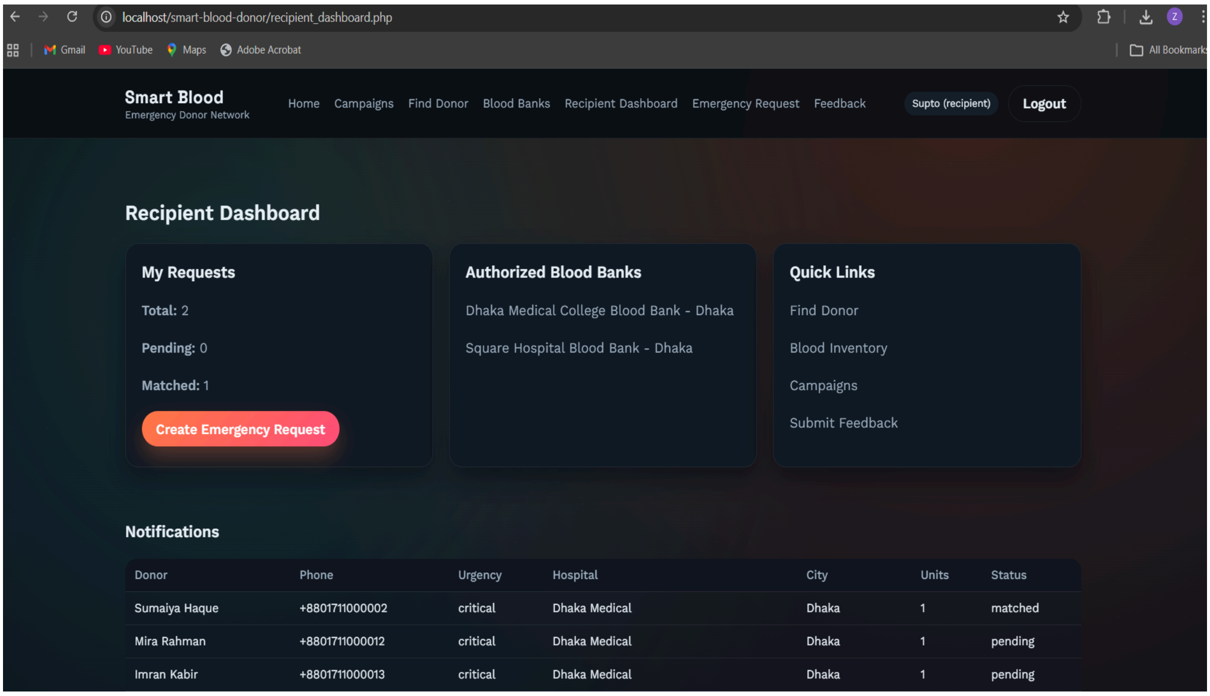
Contribution of ID : 24101339 , Name : Kazi Zarif Yamin

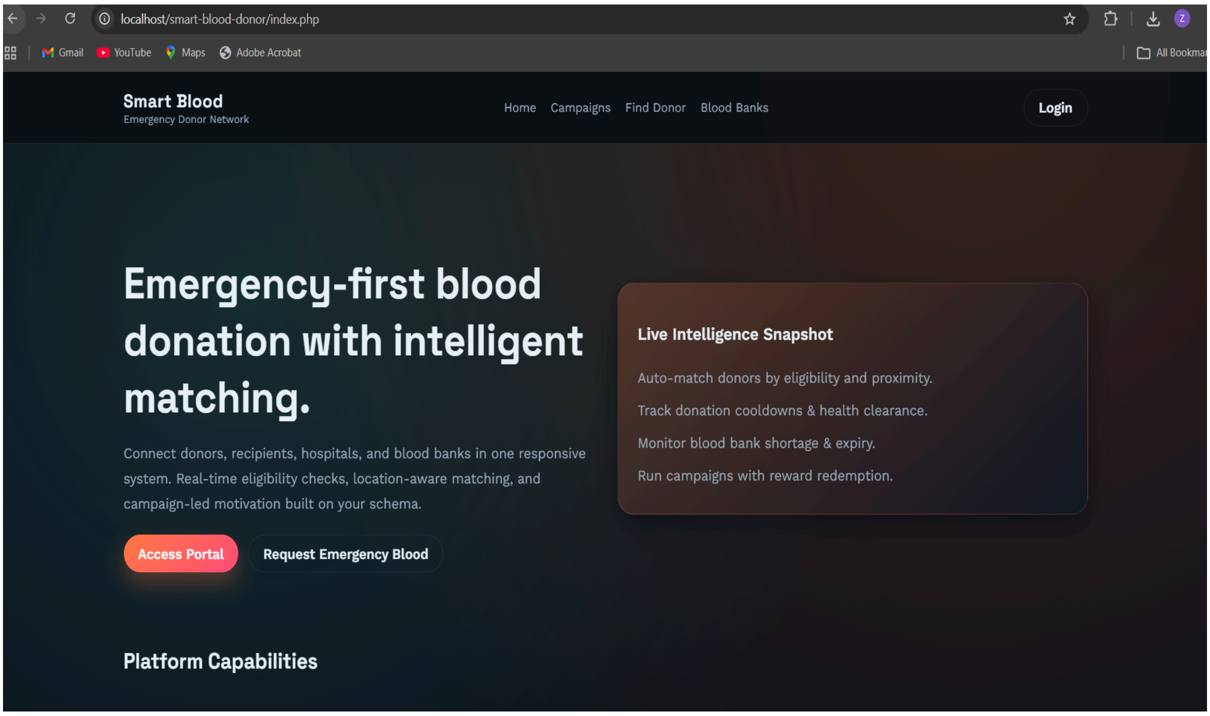
Designing clean, simplified interface elements and usability components, including the recipient dashboard statistics, straightforward feedback submission workflows and clear structured tables for the blood bank inventory presentation.

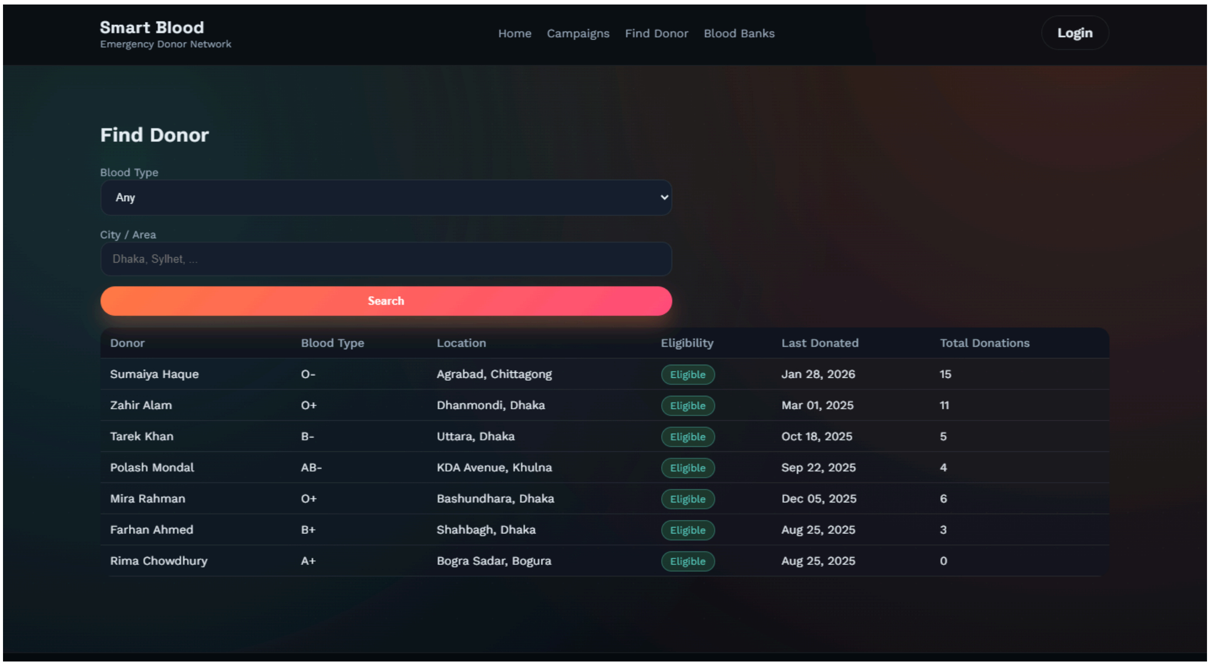
Contribution of ID : 24101363 , Name : Syeda Fatima Tasnia

Designing the core foundational layouts and user interaction pages, including the main home page structure, the global navigation bar, accessible login and registration forms, as well as the UI displays for active campaigns and reward redemption.


Smart Blood
Emergency Donor Network
Home
Campaigns
Find Donor
Blood Banks
Donor Dashboard
Masud (donor)
Logout
Donor Dashboard
Eligibility Status
Blood: B+
Last Donated: Dec 20, 2025
Next Eligible: Mar 20, 2026
Health: Review
Date Gap: Eligible
Overall: Not Eligible
View Health Card
Availability
Set availability
Available
Update
Complete eligibility checks to unlock availability.
Points & Rank
Total Donations: 13
Points: 130
Top donors appear in the leaderboard.
Health Card
Weight: 65.0 kg
Hemoglobin: 14.0 g/dL
Recorded Diseases: 1
Eligibility Rules: Weight >= 50 kg, Hemoglobin >= 13.5 g/dL, last donation gap >= 90 days.
Status: Not Eligible


localhost/smart-blood-donor/recipient_dashboard.php
Gmail
YouTube
Maps
Adobe Acrobat
All Bookmarks
Smart Blood
Emergency Donor Network
Home
Campaigns
Find Donor
Blood Banks
Recipient Dashboard
Emergency Request
Feedback
Supto (recipient)
Logout
Recipient Dashboard
My Requests
Total: 2
Pending: 0
Matched: 1
Create Emergency Request
Authorized Blood Banks
Dhaka Medical College Blood Bank - Dhaka
Square Hospital Blood Bank - Dhaka
Quick Links
Find Donor
Blood Inventory
Campaigns
Submit Feedback
Notifications
Donor
Phone
Urgency
Hospital
City
Units
Status
Sumaiya Haque
+8801711000002
critical
Dhaka Medical
Dhaka
1
matched
Mira Rahman
+8801711000012
critical
Dhaka Medical
Dhaka
1
pending
Imran Kabir
+8801711000013
critical
Dhaka Medical
Dhaka
1
pending


localhost/smart-blood-donor/index.php
Gmail
YouTube
Maps
Adobe Acrobat
All Bookma
Smart Blood
Emergency Donor Network
Home
Campaigns
Find Donor
Blood Banks
Login
Emergency-first blood donation with intelligent matching.
Connect donors, recipients, hospitals, and blood banks in one responsive system. Real-time eligibility checks, location-aware matching, and campaign-led motivation built on your schema.
Access Portal
Request Emergency Blood
Live Intelligence Snapshot
Auto-match donors by eligibility and proximity.
Track donation cooldowns & health clearance.
Monitor blood bank shortage & expiry.
Run campaigns with reward redemption.
Platform Capabilities


Smart Blood
Emergency Donor Network
Home
Campaigns
Find Donor
Blood Banks
Login
Find Donor
Blood Type
Any
City / Area
Dhaka, Sylhet, ...
Search
Donor | Blood Type | Location | Eligibility | Last Donated | Total Donations
Sumaiya Haque | O- | Agrabad, Chittagong | Eligible | Jan 28, 2026 | 15
Zahir Alam | O+ | Dhanmondi, Dhaka | Eligible | Mar 01, 2025 | 11
Tarek Khan | B- | Uttara, Dhaka | Eligible | Oct 18, 2025 | 5
Polash Mondal | AB- | KDA Avenue, Khulna | Eligible | Sep 22, 2025 | 4
Mira Rahman | O+ | Bashundhara, Dhaka | Eligible | Dec 05, 2025 | 6
Farhan Ahmed | B+ | Shahbagh, Dhaka | Eligible | Aug 25, 2025 | 3
Rima Chowdhury | A+ | Bogra Sadar, Bogura | Eligible | Aug 25, 2025 | 0

Backend Development

The backend development was done using PHP and MySQL.

Contribution of ID : 24101406, Name : Farhan Sadik

Development of emergency request processing logic, including database insertion, input validation, automated notification signals, dynamic donor health and points profile updates, calculation of 90-day eligibility intervals and implementation of the find donor compatibility and multi-condition search backend.

Contribution of ID : 24101339 , Name : Kazi Zarif Yamin

Development of feedback data operations (managing submissions, ratings, and comments tied to donation trackers), recipient dashboard data retrieval and validation for ongoing and fulfilled requests and blood bank inventory operations including stock updates, shortage alerts, and expiration date checks.

Contribution of ID : 24101363 , Name : Syeda Fatima Tasnia

Development of reward claim database transactions ensuring data consistency, validation of reward availability, active campaign management logic, and admin moderation endpoints for monitoring requests, campaigns and verifying user actions.

config.php

app > config.php

```php
<?php
// Basic app configuration. Update DB credentials to match your MySQL setup.
return [
    'db' => [
        'host' => 'localhost',
        'user' => 'root',
        'pass' => '',
        'name' => 'smart_blood_donor',
        'charset' => 'utf8mb4',
        'port' => 3307
    ],
    'app_name' => 'Smart Blood & Emergency Donor Network',
    'timezone' => 'Asia/Dhaka',
];
```

db.php

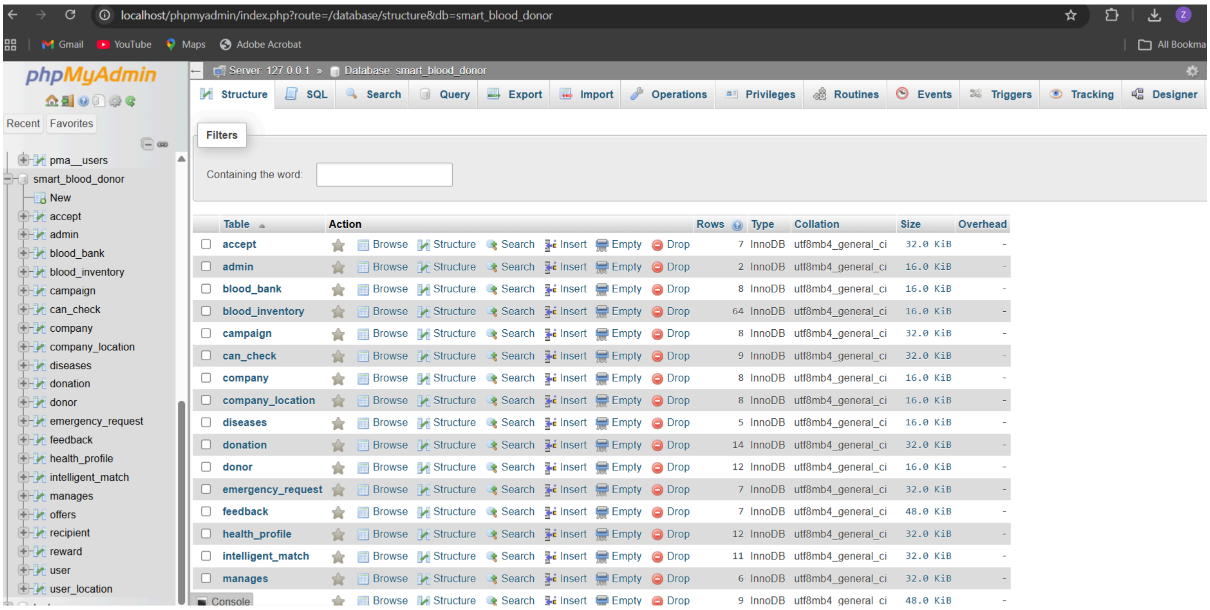
app > db.php

```php
<?php
$config = require __DIR__ . '/config.php';
date_default_timezone_set($config['timezone']);

$mysqli = new mysqli(
    $config['db']['host'],
    $config['db']['user'],
    $config['db']['pass'],
    $config['db']['name'],
    $config['db']['port']
);

if ($mysqli->connect_errno) {
    die('Database connection failed: ' . $mysqli->connect_error);
}

$mysqli->set_charset($config['db']['charset']);
```


localhost/phpmyadmin/index.php?route=/database/structure&db=smart_blood_donor
Gmail
YouTube
Maps
Adobe Acrobat
All Bookma
phpMyAdmin
Recent
Favorites
pma__users
smart_blood_donor
New
accept
admin
blood_bank
blood_inventory
campaign
can_check
company
company_location
diseases
donation
donor
emergency_request
feedback
health_profile
intelligent_match
manages
offers
recipient
reward
user
user_location
Server: 127.0.0.1 » Database: smart_blood_donor
Structure
SQL
Search
Query
Export
Import
Operations
Privileges
Routines
Events
Triggers
Tracking
Designer
Filters
Containing the word:
Table
Action
Rows
Type
Collation
Size
Overhead
accept Browse Structure Search Insert Empty Drop 7 InnoDB utf8mb4_general_ci 32.0 KiB -
admin Browse Structure Search Insert Empty Drop 2 InnoDB utf8mb4_general_ci 16.0 KiB -
blood_bank Browse Structure Search Insert Empty Drop 8 InnoDB utf8mb4_general_ci 16.0 KiB -
blood_inventory Browse Structure Search Insert Empty Drop 64 InnoDB utf8mb4_general_ci 16.0 KiB -
campaign Browse Structure Search Insert Empty Drop 8 InnoDB utf8mb4_general_ci 32.0 KiB -
can_check Browse Structure Search Insert Empty Drop 9 InnoDB utf8mb4_general_ci 32.0 KiB -
company Browse Structure Search Insert Empty Drop 8 InnoDB utf8mb4_general_ci 16.0 KiB -
company_location Browse Structure Search Insert Empty Drop 8 InnoDB utf8mb4_general_ci 16.0 KiB -
diseases Browse Structure Search Insert Empty Drop 5 InnoDB utf8mb4_general_ci 16.0 KiB -
donation Browse Structure Search Insert Empty Drop 14 InnoDB utf8mb4_general_ci 32.0 KiB -
donor Browse Structure Search Insert Empty Drop 12 InnoDB utf8mb4_general_ci 16.0 KiB -
emergency_request Browse Structure Search Insert Empty Drop 7 InnoDB utf8mb4_general_ci 32.0 KiB -
feedback Browse Structure Search Insert Empty Drop 7 InnoDB utf8mb4_general_ci 48.0 KiB -
health_profile Browse Structure Search Insert Empty Drop 12 InnoDB utf8mb4_general_ci 32.0 KiB -
intelligent_match Browse Structure Search Insert Empty Drop 11 InnoDB utf8mb4_general_ci 32.0 KiB -
manages Browse Structure Search Insert Empty Drop 6 InnoDB utf8mb4_general_ci 32.0 KiB -
Console Browse Structure Search Insert Empty Drop 9 InnoDB utf8mb4_general_ci 48.0 KiB -

Source Code Repository

GITHUB: https://github.com/F3uR0n/smart-blood-donor-system

Conclusion

In conclusion, Smart Blood & Emergency Donor Network is a structured and practical database system that solves a real and urgent problem connecting blood donors with recipients quickly and safely. Through role-based access for donors, recipients, admins and a clear request-to-donation workflow. The platform keeps every interaction organized and easy to follow.

The intelligent matching engine is the heart of the system. It checks blood type, weight, hemoglobin, disease history, 90-day cooldown and location proximity before suggesting any donor ensuring every match is both medically safe and realistic.

Beyond emergency handling, the system includes campaigns with reward redemption to motivate repeat donors, a leaderboard to encourage healthy competition, a feedback system for recipients to rate donors, and a blood bank inventory viewer. Together, these features make the platform complete.

Overall, our project shows how a well-designed relational database can manage complex, multi-role workflows involving real-time status updates, conditional eligibility logic and interconnected entities. The Smart Blood & Emergency Donor Network is a meaningful DBMS project with direct real-world relevance and a clear potential for actual deployment.

References

1. https://www.youtube.com/watch?v=d5Hf6d6QtIo
2. https://www.youtube.com/watch?v=J8QPm8_j6lM